

RETRIEVAL-AUGMENTED GENERATION

30 Multiple-Choice Questions for Freshers

Questions • Four Options • Correct Answers • Simple Justifications

Topics Covered

RAG basics, document loading, chunking, tokens, embeddings, vector databases, semantic search, Top-K retrieval, context, metadata, sources, hallucination control, re-ranking, conversation memory, and document deletion.

How to use

First choose an answer without looking at the solution. Then read the correct answer and justification to understand the concept.

1. What does RAG stand for?

- A. Rapid AI Generation
- B. Retrieval-Augmented Generation
- C. Random Answer Generator
- D. Response Analysis Graph

Correct Answer: B. Retrieval-Augmented Generation

Justification: RAG stands for Retrieval-Augmented Generation. It retrieves relevant information first and then uses that information to generate an answer.

2. What is the main purpose of RAG?

- A. To increase computer memory
- B. To generate answers using relevant external knowledge
- C. To replace all databases
- D. To design web pages

Correct Answer: B. To generate answers using relevant external knowledge

Justification: RAG helps an LLM answer with information retrieved from documents or other knowledge sources instead of depending only on its training data.

3. Which sequence correctly represents a basic RAG pipeline?

- A. Question - Retrieval - Context - LLM - Answer
- B. Answer - Database - Question - LLM
- C. LLM - Delete - Upload - Answer
- D. Question - CSS - HTML - Answer

Correct Answer: A. Question - Retrieval - Context - LLM - Answer

Justification: A RAG system receives a question, retrieves relevant content, gives it to the LLM as context, and then generates the answer.

4. What is document loading in RAG?

- A. Extracting content from files
- B. Deleting documents
- C. Creating passwords
- D. Changing the screen resolution

Correct Answer: A. Extracting content from files

Justification: Document loading reads and extracts text or other content from sources such as PDF, DOCX, TXT, or web pages.

5. Why is a large document divided into chunks?

- A. To make the file colourful
- B. To create smaller searchable pieces
- C. To remove all text
- D. To convert it into an image

Correct Answer: B. To create smaller searchable pieces

Justification: Smaller chunks are easier to embed, search, retrieve, and fit inside the LLM's context window.

6. What is a chunk in RAG?

- A. A small section of document content
- B. A database password
- C. A complete operating system
- D. An API error

Correct Answer: A. A small section of document content

Justification: A chunk is a manageable piece of a larger document, often containing a paragraph or a limited number of characters or tokens.

7. What is chunk overlap?

- A. Repeated content shared between neighbouring chunks
- B. Deleting duplicate files
- C. Combining two databases
- D. Using two LLMs

Correct Answer: A. Repeated content shared between neighbouring chunks

Justification: Overlap repeats a small amount of text between adjacent chunks so that important meaning is not lost at chunk boundaries.

8. What is a token?

- A. A unit of text processed by a language model
- B. A PDF page
- C. A database table
- D. A computer monitor

Correct Answer: A. A unit of text processed by a language model

Justification: A token is a small unit of text, such as a word, part of a word, or punctuation, that an AI model processes.

9. What is an embedding?

- A. A numerical vector representing meaning
- B. A type of keyboard
- C. A CSS property
- D. A PDF password

Correct Answer: A. A numerical vector representing meaning

Justification: An embedding converts text into a list of numbers that captures semantic meaning and enables similarity comparison.

10. Which model is used to convert text into vectors?

- A. Embedding model
- B. Image editor
- C. Web browser
- D. SQL compiler

Correct Answer: A. Embedding model

Justification: An embedding model converts documents and questions into vectors in the same numerical space.

11. Where are embeddings commonly stored?

- A. Vector database
- B. HTML table only
- C. Operating-system folder only
- D. Email inbox

Correct Answer: A. Vector database

Justification: A vector database stores embeddings and supports fast similarity searches to find semantically related content.

12. Which of the following is a vector database or vector store?

- A. ChromaDB
- B. Microsoft Word
- C. PowerPoint
- D. Notepad

Correct Answer: A. ChromaDB

Justification: ChromaDB is designed to store embeddings, metadata, and documents and to perform vector similarity searches.

13. What does FAISS mainly provide?

- A. Efficient similarity search over vectors
- B. Email delivery
- C. PDF editing
- D. Frontend styling

Correct Answer: A. Efficient similarity search over vectors

Justification: FAISS is a library used to index vectors and quickly find the vectors closest to a query vector.

14. What is semantic search?

- A. Search based on meaning
- B. Search based only on file size
- C. Search based only on colour
- D. Search for computer viruses

Correct Answer: A. Search based on meaning

Justification: Semantic search compares meanings represented by embeddings, so it can find relevant content even when the exact words differ.

15. How is a user's question used during retrieval?

- A. It is converted into an embedding
- B. It is converted into CSS
- C. It is permanently deleted
- D. It is used as a password

Correct Answer: A. It is converted into an embedding

Justification: The question is embedded using the same embedding model so its vector can be compared with document vectors.

16. What is similarity search?

- A. Finding document vectors closest to the question vector
- B. Checking two file names only
- C. Sorting users by age
- D. Searching the internet speed

Correct Answer: A. Finding document vectors closest to the question vector

Justification: Similarity search measures how close the question embedding is to stored document embeddings and returns the closest matches.

17. What does Top-K mean in RAG?

- A. The number of most relevant chunks retrieved
- B. The number of users
- C. The LLM password length
- D. The document font size

Correct Answer: A. The number of most relevant chunks retrieved

Justification: Top-K controls how many of the highest-scoring chunks are returned from similarity search.

18. If Top-K is set to 3, what is normally retrieved?

- A. The three most relevant chunks
- B. Three complete databases
- C. Three LLM models
- D. Three user passwords

Correct Answer: A. The three most relevant chunks

Justification: Top-K equal to 3 requests the three chunks with the best similarity scores.

19. What is retrieved context?

- A. Relevant information supplied to the LLM
- B. The user's computer settings
- C. An empty database
- D. A website colour theme

Correct Answer: A. Relevant information supplied to the LLM

Justification: Retrieved context is the selected document content included in the prompt to help the LLM answer accurately.

20. What is the role of the LLM in a RAG system?

- A. Generate an answer using the question and retrieved context
- B. Store files permanently
- C. Create database hardware
- D. Scan the keyboard

Correct Answer: A. Generate an answer using the question and retrieved context

Justification: The LLM reads the user's question together with retrieved context and produces a natural-language response.

21. What is metadata in a RAG system?

- A. Information such as filename, page number, or category
- B. Only the document text
- C. A computer game
- D. An internet password

Correct Answer: A. Information such as filename, page number, or category

Justification: Metadata describes a chunk and helps identify, cite, filter, or manage its source.

22. Why should page number and filename be stored as metadata?

- A. To show the source of an answer
- B. To increase screen brightness
- C. To change the LLM temperature
- D. To install Python

Correct Answer: A. To show the source of an answer

Justification: Source metadata allows the application to show users where the retrieved information came from.

23. What is metadata filtering?

- A. Restricting retrieval by fields such as filename or category
- B. Changing text colour
- C. Removing every embedding
- D. Sorting HTML elements

Correct Answer: A. Restricting retrieval by fields such as filename or category

Justification: Metadata filtering narrows the search to chunks that satisfy conditions, such as belonging to a selected document or category.

24. How can RAG help reduce hallucination?

- A. By grounding answers in retrieved evidence
- B. By making the model guess more
- C. By removing all context
- D. By increasing randomness

Correct Answer: A. By grounding answers in retrieved evidence

Justification: RAG provides relevant evidence to the LLM, reducing the need to invent unsupported facts. It reduces but does not completely eliminate hallucinations.

25. Which instruction best prevents answers outside the retrieved context?

- A. Answer creatively even when information is missing
- B. Use only the provided context; otherwise say the answer is not available
- C. Always guess the answer
- D. Ignore all retrieved documents

Correct Answer: B. Use only the provided context; otherwise say the answer is not available

Justification: This instruction grounds the response in retrieved evidence and tells the model what to do when the context does not contain the answer.

26. What does a similarity score indicate?

- A. How closely a retrieved chunk matches the query
- B. How long the filename is
- C. How many users are logged in
- D. How colourful the document is

Correct Answer: A. How closely a retrieved chunk matches the query

Justification: A similarity score represents closeness between the query vector and a stored chunk vector, although the exact scale depends on the method used.

27. What is re-ranking in RAG?

- A. Reordering retrieved results by relevance
- B. Renaming PDF files
- C. Deleting the vector database
- D. Increasing screen size

Correct Answer: A. Reordering retrieved results by relevance

Justification: A re-ranker evaluates initially retrieved chunks more carefully and places the most relevant ones first.

28. What is conversation memory in a RAG chatbot?

- A. Stored chat history used to understand follow-up questions
- B. The computer's RAM only
- C. A PDF bookmark
- D. A database password

Correct Answer: A. Stored chat history used to understand follow-up questions

Justification: Conversation memory keeps earlier messages or a summary so the chatbot can understand references and follow-up questions.

29. What should happen when a document is deleted from a RAG application?

- A. Its document record, chunks, and embeddings should be removed
- B. Only the screen should refresh
- C. The LLM should be uninstalled
- D. Every user's data should be deleted

Correct Answer: A. Its document record, chunks, and embeddings should be removed

Justification: Removing all related records prevents deleted content from appearing in future retrieval results.

30. Which formula best summarizes RAG?

- A. User Question + Retrieved Context + LLM = Grounded Answer
- B. HTML + CSS = Vector
- C. Password + Database = Prompt
- D. PDF + Colour = LLM

Correct Answer: A. User Question + Retrieved Context + LLM = Grounded Answer

Justification: RAG combines the user's question with relevant retrieved information and sends both to an LLM to produce a grounded answer.